

Chit 2 — PL/SQL: Library Fine System

Table Creation

```
CREATE TABLE Borrower (  
  Rollin    NUMBER PRIMARY KEY,  
  Name      VARCHAR2(50),  
  DateofIssue DATE,  
  NameofBook VARCHAR2(100),  
  Status    CHAR(1)  
);  
  
CREATE TABLE Fine (  
  Roll_no NUMBER,  
  Date    DATE,  
  Amt     NUMBER(10,2)  
);  
  
-- Sample data  
INSERT INTO Borrower VALUES (101, 'Amit', SYSDATE-35, 'DBMS', 'I');  
INSERT INTO Borrower VALUES (102, 'Sneha', SYSDATE-20, 'OS', 'I');  
COMMIT;
```

PL/SQL Block

```
DECLARE  
  v_roll    Borrower.Rollin%TYPE := &roll_no;  
  v_book    Borrower.NameofBook%TYPE := '&book_name';  
  v_doi     DATE;  
  v_days    NUMBER;  
  v_fine    NUMBER := 0;  
BEGIN  
  -- Fetch issue date  
  SELECT DateofIssue INTO v_doi  
  FROM Borrower  
  WHERE Rollin = v_roll AND NameofBook = v_book AND Status = 'I';  
  
  v_days := TRUNC(SYSDATE - v_doi);  
  
  -- Fine Calculation  
  IF v_days BETWEEN 15 AND 30 THEN  
    v_fine := v_days * 5;  
  ELSIF v_days > 30 THEN  
    v_fine := (30 * 5) + ((v_days - 30) * 50);  
  END IF;  
  
  -- Update status to Returned  
  UPDATE Borrower  
  SET Status = 'R'  
  WHERE Rollin = v_roll AND NameofBook = v_book;  
  
  -- Insert fine if applicable  
  IF v_fine > 0 THEN  
    INSERT INTO Fine VALUES (v_roll, SYSDATE, v_fine);  
    DBMS_OUTPUT.PUT_LINE('Fine of Rs ' || v_fine || ' imposed.');  ELSE  
    DBMS_OUTPUT.PUT_LINE('No fine. Book returned within 15 days.');  END IF;
```

```
    COMMIT;
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Error: Record not found for given Roll No and Book.');
```

When Others then

```
        DBMS_OUTPUT.PUT_LINE('Unexpected Error: ' || SQLERRM);
END;
/
```

Note: Days 1-14: No fine. Days 15-30: Rs 5/day. Days >30: Rs 5/day for first 30 + Rs 50/day beyond 30.